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University of Salahaddin-Erbil
College of Engineering
Department of Software and Informatics

Similarity And Classification Of Prophet Hadiths ﷺ Muhammad

A Project Submitted to the Department of Software and Informatics Engineering
University of Salahaddin-Erbil
In the Partial Fulfillment of the Requirement for the Degree of Bachelor of Science
in Software and Informatics Engineering

Prepared by:

Nuraddin Hassan Rasul

Ali Farhad Ali

Project Supervised by:

M. Sanar Fawzi

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ABSTRACT

The objective of this project is to develop a software solution that clusters and classifies Hadiths based on their similarity in Matn (textual content). The clustering and classification of Hadiths play a vital role in organizing and making the vast corpus of Hadiths more accessible and navigable for users, including websites, Hadith science specialists, and other individuals interested in the field.

The primary focus of the project is not only on the generation of clustered and classified data, but also on the creation of a software solution that is extensible and maintainable. This is essential because the initial algorithms used for clustering and classification may not yield highly accurate results. Therefore, the project adopts an iterative approach, allowing for refinement and improvement of the algorithms and techniques employed.

To accomplish the objectives of the project, a combination of software technologies and data analysis methods are utilized. In particular, concepts from the field of Natural Language Processing (NLP) are applied to effectively analyze and process the textual content of the Hadiths. However, it is important to consider the challenges and limitations posed by the Arabic language, such as its complex grammar and syntax, which may require specialized tools or SDKs (Software Development Kits).

The results of the project will provide insights into the accuracy achieved for both the clustering and classification tasks. Additionally, the project will contribute to the development of tools and techniques that can be further refined and enhanced in the future to improve the accuracy of the clustering and classification algorithms. By creating a software solution that is flexible and adaptable, it becomes possible to incorporate future advancements in the field, ensuring the continuous improvement of the system.

DEDICATION

We are dedicating our work to our beloved families who have stood by us for all our lives and raised us the best way they could. Our teachers and supervisor who taught us and made it possible for us to achieve such big projects. And our dear friends who have been there for us when we needed them and have always supported us.

SUPERVISOR CERTIFICATE

I certify that the engineering project titled "similarity and classification of Prophet Muhammad ﷺ hadiths" was done under my supervision at the Department of Software and Informatics Engineering, College of Engineering, Salahaddin University in Erbil. In the partial fulfillment of 'The requirement for the degree of Bachelor of Science in Softwareand Informatics Engineering'.

Supervisor

Signature:

Name: Sanar Fawzi

EXAMINER CERTIFICATE

I certify that I have read and examined this project titled “similarity and classification of Prophet Muhammad ﷺ hadiths" and as an examining committee member I have examined the students:

- Nuraddin Hassan Rasul
- Ali Farhad Ali

And it meets the standard of a report for the degree of Bachelor of Science in Software and Informatics Engineering’.

Committee Member

Signature:

Name: Silva Hekmat

Date: / /2023

Committee Member

Signature:

Name: Dilnya Sabir

Date: / /2023

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LIST OF ABBREVIATIONS

TF-IDF: Term Frequency-Inverse Document Frequency

NLP: Natural Language Processing

API: Application Programming Interface

HTML: Hypertext Markup Language

CSS: Cascading Style Sheets

SDK: Software Development Kit

PHP: Hypertext Preprocessor

RDBMS: Relational Database Management System

IDE: Integrated Development Environment.

1. INTRODUCTION

1.1 BACKGROUND

The study and interpretation of Hadiths, the sayings and actions of Prophet Muhammad ﷺ, hold significant importance in Islamic scholarship. Hadiths provide guidance and insights into various aspects of religious teachings, moral values, and legal principles. As the number of available Hadiths has increased between books, the need for effective organization, analysis, and classification of this vast corpus has become evident.one of main fields of hadith science is called Takhrij (تخـــريـــج) that mean know the Shawahid (شـــواهـــد) those similar as evidence of hadith and determining the books those contain them. And also contain Tabwib (تـــبويـــب) that represent the Classification of hadith. (Azami, 1978)

The arrival of technology, particularly in the field of Natural Language Processing (NLP) and machine learning, has opened new possibilities for the systematic analysis of Hadiths. By leveraging computational methods, it becomes feasible to categorize and cluster Hadiths based on their themes, topics, or narrators. Such classification and clustering can aid scholars, researchers, and general readers in accessing relevant Hadiths and studying them within specific contexts.

The project is collecting a comprehensive dataset of Hadiths and apply machine learning techniques to cluster and classify them based on shared characteristics. By employing a data-driven approach, we seek to uncover patterns and relationships within the Hadith corpus.

The process of clustering and classification in this project will be on Matn section of Hadith. But first brief understanding of Hadith structure. Hadith include two sections, Hadith start with Sanad (سند) that is chain of narrators and subsequent

section is Matn (متن) the contained of hadith that include the words or actions attributed to the prophet Mohammed ﷺ.

The background of tech or software's have done these aspects in Takhrij science was lack of two certain things.

- First: availability of all books.
- Second: The accuracy of the results obtained.

To address these challenges, our approach involves employing NLP (Natural Language Processing) technology in conjunction with the expertise and guidance of Hadith scholars. By leveraging advanced technology, we aim to overcome the obstacle of manually examining over 1000 books. This collaborative effort between technology and scholars will enhance the availability and accuracy of the Takhrij process.

This report outlines the methodology, processes, and outcomes of our project, providing insights into the data collection, clustering algorithms employed, classification techniques utilized, and the significance of our findings. By exploring the potential of machine learning in the context of Hadith analysis, we hope to contribute to the broader field of Islamic studies and provide a valuable resource for scholars and practitioners.

1.2 PROBLEM STATEMENT

The study and analysis of Hadiths, the sayings and actions of Prophet Muhammad, are of great significance in Islamic scholarship. However, the vast amount of available Hadith literature poses several challenges in effectively organizing, categorizing, and accessing this valuable information. The following are the key problems that our project aims to address:

1. **Limited Accessibility**: Traditional methods of accessing Hadiths often involve manual searches through printed books or online repositories, which can be time-consuming and inefficient. There is a need for a more accessible and user-friendly system that allows researchers, scholars, and the general public to search and explore Hadiths based on specific criteria or topics.

2. **Unstructured Data** : Hadiths are available in various formats and translations, making it difficult to extract and organize the relevant information. Though standardized structures of hadiths books that scholars achieved, there is The lack of united structures and metadata associated with Hadiths hampers efficient retrieval and analysis.

3. **Manual Categorization:** The process of categorizing Hadiths traditionally relies on manual effort, leading to subjective interpretations, inconsistencies, and labor-intensive tasks. Automating the categorization process using machine learning techniques can streamline this process and provide more objective and consistent results.

4. **Knowledge Discovery:** By applying machine learning algorithms to analyze and cluster Hadiths, there is an opportunity to uncover hidden patterns, relationships, and insights within the vast corpus. However, there is a lack of systematic tools and approaches that harness the power of machine learning for knowledge discovery in Hadith literature.

5. even the achievement in software industry in this field, still they lack a lot of implementation issues as including all books or other issues

Addressing these challenges will contribute to the development of a more accessible, organized, and efficient system for exploring and analyzing Hadiths. By leveraging technology, particularly machine learning techniques, we aim to overcome the limitations of traditional methods and provide valuable resources for researchers, scholars, and anyone interested in studying the teachings and principles of Prophet Muhammad as captured in Hadiths.

1. 3 AIMS AND OBJECTIVES

The aim of our project is to develop a comprehensive system for collecting and organizing hadiths data. Then clustering and categorizing Hadiths of Prophet Muhammad using machine learning techniques. This system will provide a valuable resource for researchers, scholars, and individuals interested in studying and understanding the teachings and principles advocated by Prophet Muhammad.To accomplish this aim, we have defined the following objectives:

1. **Collecting Hadiths:** Gather a diverse and representative dataset of authentic Hadiths from reliable sources, ensuring accuracy and adherence to scholarly standards.

2. **Preprocessing and Text Representation software:** build software for Applying preprocessing techniques to clean and normalize the Hadith texts, removing noise and irrelevant information. That mean transform the textual hadiths into suitable representations for machine learning algorithms as numerical features. such as word embedding or BOW algorithms

3. **Data Clustering:** Utilize clustering algorithms AI models, such as k-means or hierarchical clustering, to group similar Hadiths together based on textual similarities. The clustering process will enable the identification of common themes, topics, or contextual relationships within the Hadith corpus.
4. **Data Classification**: Develop classification models, such as supervised machine learning algorithms or rule-based approaches, to assign categories or topics to the clustered Hadiths. The classification will provide a structured organization of the Hadiths, facilitating easier retrieval and navigation based on specific subjects.
5. **Hadith checker web application:** Design and implement an intuitive user interface that allows researchers and scholars to search, browse, and access the categorized Hadiths. The user interface should provide a seamless and interactive experience, facilitating efficient exploration and analysis of the Hadith corpus.
6. **Evaluation algorithms:** Assess the performance and effectiveness of the clustering and classification algorithms used in the system. Conduct quantitative and qualitative evaluations to measure the accuracy, efficiency, and user satisfaction of the system's clustering and categorization capabilities.

By accomplishing these objectives, we aim to create a robust system that contributes to the organization, analysis, and accessibility of Hadiths, empowering researchers and scholars in their study of Prophet Muhammad's teachings and promoting a deeper understanding of Islamic principles.

1.4 CHAPTER OUTLINES

Chapter 2 presents the methodology, explaining the iterative process model used and the stages involved, including requirement gathering, analysis, software design, implementation, and testing. Chapter 3 focuses on the analysis and design phase, covering the analysis of collected Hadiths and the design of clustering and classification algorithms, along with data preprocessing techniques and feature selection. In Chapter 4, the results and discussions are presented, showcasing the outcomes of the clustering and classification algorithms, analyzing their performance, accuracy, and addressing any challenges or limitations faced. The final chapter is a place where the conclusion and future works are all documented.

2. METHODOLOGY

Fortunately, we were able to collects a huge number of Hadiths through so many techniques and we found many API for many Hadith books. We can say that we had all the Hadiths we want to work with but also it was very challenging at the same time we had to organize that big data that includes testing, clean and analyzing it constantly and this leads us to choose a well-suited software model for the project's development process.

Iterative models are commonly used in data analysis and machine learning tasks, where the initial results may not be optimal, and refinement is necessary. By iterating through the process, you likely made adjustments to parameters, experimented with different techniques, and refined the clustering and classification algorithms to improve the accuracy and quality of the results.

2.1 Iterative Software Model

The iterative process model is a rework scheduling strategy in which time is set aside to revise and improve parts of the system. In this case, feature code is designed, developed and tested in repeated cycles. A detailed requirement is not required to start with as improvement starts by indicating and executing parts of the software. This procedure is then rehashed to create another rendition of the software toward the finish of every cycle of the model.

Iterative development model separates the product advancement of an expansive application into little pieces. It is therefore regarded as the process of breaking down the software development of a large application into smaller chunks. In our case we knew our requirements but the problem was the data was not clean and we had to iterate so many times until we get more clean data so that our classification and clustering will become more accurate.

The iterative model was designed as an improvement to the existing waterfall model. The waterfall model is a linear SDLC model whereas the iterative model is cyclical in nature.

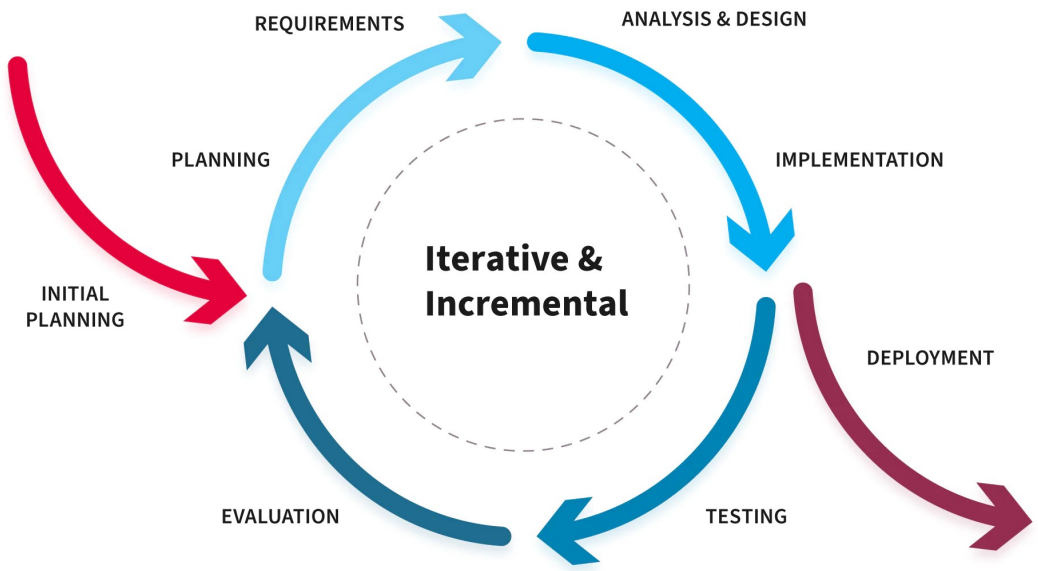

Figure 1 stages of Iterative model

2.2 REQUIREMENT ANALYSIS

2.2.1 Perspective

identifying and classifying hadith is more like a text docs data analyzing project that aims to serve Islamic religion in a way that it works and analyze hadiths of prophet Mohammad ﷺ for researchers and normal users and also it serves as source of other projects.

This project in general includes these parts:

- Algorithms for manipulating data and applying cluster and classification
- A web application, designed as an admin dashboard, is created specifically for researchers and scholars. This web app provides a user-friendly interface that allows researchers to access and explore the Hadith database. Researchers can search for specific Hadiths, browse through categorized Hadiths, and have the ability to make corrections or updates to the data if any mistakes are found.
- A robust and efficient database system is an integral part of the project. The database is responsible for storing and managing the collected Hadiths and associated metadata. It provides a reliable and scalable foundation to support the storage and retrieval of data for the clustering, classification, and research purposes.

2.2.2 Functional Requirement

- **Data Collection:** Specify the requirements for collecting Hadiths data from authentic sources. Determine the sources to be used, such as well-known Hadith books or databases, and define the process for gathering the Hadith texts.

- **Clustering:** Clustering hadiths according to similarity in Matn, using word-vectored feature extracted by considering the importance and position of words. Accuracy of resulted clustered is determined by hadiths similarity included in a cluster not by the percentage of hadith those being clustered. But at least about 110,000 hadiths should be clustered.
- **Classification:** Determine the classes of hadiths. According to the similarity of hadiths and chapters of books.
- **Reusable Prepared data**

  Data that working on belongs to those four categories.
  - Hadiths brought from books
  - Cleaned hadith Matn separated from Sanad
  - Preprocessed Matn to be ready for feature extraction
  - Feature extracted data
  - Clustered and classified data. As they will be needed to compare with new resulted clustering

- **Collaboration with hadith specialists**

  build web app that enable hadith specialists to contract with hadith for:
  - check hadith content against the original in pdf format.
  - check cleaned Matn separated from Sanad of hadith
  - evaluate the position of hadith in a book, it's number, chapter and subchapter associated with

2.2.3 Non-Functional Requirement

- **Performance**

  as dealing with a lot of hadith data, fetching from database using, preprocessing them and do feature extractions. The tools, packages, data structures and AI models should have fast performance but not at the expense of accuracy in results.

- **Reliability**

  is an important Non-Functional Requirement for most software products, the system must work without failure at any time, making sure that the system is reliable in its operations and for securing the sensitive details.

- **No Redundancy**

  In the proposed system you must ensure that no information is repeated anywhere, in storage or otherwise. This would assure economic use of storage space and consistency in the data stored.

- **Maintainability**

  Because of the repetition of go throw the life cycle of development, care must be taken to ensure that the code is adjustable and extensible.

- **Security**

  Keep database secured for unauthenticated access, and ensure authentication and authorization of web application that provide to hadith specialist checkers

2.3 IMPLEMENTATION

Software and Tools that are needed to implement this project include:

IDE:

- **PHPstorm**: free version for student community that enable great experience when working with PHP and it's farmwards, Visual Studio Code: these 2 are one of the best IDE for any kind of software development.
- **VSCode**: Using VS Code for Python development offers several advantages. It is a lightweight and fast editor with extensive customization options through various extensions. The integrated terminal allows for easy script execution, while intelligent code completion enhances productivity. The powerful debugging capabilities enable efficient troubleshooting, and seamless integration with version control systems simplifies collaboration. With cross-platform support and a thriving community, VS Code is a popular choice for Python developers seeking a productive and versatile coding environment.

Programming languages:

- **python**: as our project has open flexible future for using tech then Python was chosen because it arose and become as popular and prime choice for (NLP) due to its versatility and extensive capabilities in addressing analyzing text data. Its intuitive simple syntax and readability make it an accessible language for NLP tasks, enabling specialists and developers to implement multiple techniques like tokenization, stemming, part-of-speech tagging, named entity recognition, and sentiment analysis. Python's extensive community support and active development ecosystem contribute to its success in the NLP field, as it allows for the sharing of knowledge, resources, and pre-trained models. With Python, NLP practitioners can effectively analyze and process text data, opening up opportunities for applications in areas such as machine translation, chatbots, text summarization, and information extraction. (pythoncentral.io, 2023)

- **PHP**: is a popular choice for web application development due to its simplicity, wide community support, and broad compatibility with different web servers and operating systems.

Laravel framework: is a popular PHP framework known for its simplicity and extensive features. It offers a robust ecosystem, including features like database management, routing, and modular architecture. With Laravel, developers can build secure and scalable web applications efficiently. Its active community and comprehensive documentation make it a preferred choice for PHP web development.

Scikit-learn library: is a widely-used Python library for machine learning. It provides a range of tools and algorithms for tasks such as classification, regression, clustering, and dimensionality reduction.

Why scikit-learn:

1. As we with fresh experience with AI concepts and models developers can easily implement machine learning models and perform data preprocessing
2. Customizable as it is open source
3. Has the most Powerful models for docs clustering
4. Rich community

MiniBatchKMeans : clustering machine leaning model is a variant of the KMeans clustering algorithm that is designed to handle large datasets efficiently. It is particularly useful when working with datasets that do not fit entirely in memory. even that KMeans has advantage of accuracy rate comparing to MiniBatchKMeans but not able to implement duo to the big data corpus the database

LogisticRegression: clustering machine leaning model, it provides interpretable results, allowing us to understand the impact of features on predictions. The algorithm is computationally efficient, making it suitable for large datasets as our data. Logistic Regression is robust to noise, performs well with limited data, and has a lower risk of overfitting. It is highly scalable and can handle high-dimensional datasets effectively. Additionally, Logistic Regression is easy to implement, requiring minimal hyperparameter tuning. Overall, it strikes a good balance between

interpretability and performance, making it a popular choice in machine learning tasks.

PostgreSQL : As working on project of clustering and classification in addition even in future do word similarity, we at least need a data repository that should preserve resource of data and be able to handle access, insert, modify of them and build relations between them.

And because we have large amount of data about 450,000 of hadiths, one of best database for That was PostgreSQL that is RDBMS that has features of:

1. Has array datatype as text[] that being used extensively in database.
2. Robust performance especially according to speed factor
3. Active community of developer
4. Low cast
5. Customizable as customizing full-text search tool

Pandas: as working with large amount of data, python original list data structure. Is slow, so we select pandas as main data structure to deal with list

Arabert/Farasapy and **Arabic-Stopwords**: Arabert/Farasapy it is a powerful library to do segmenting and stemming in Arabic language and Arabic-Stopwords package that provide large range of words but lack of modifying, these are all very useful tools for preprocessing data stage most of them was part of (NLP)

Frontend tools:

Html: a markup language used for structuring web content. It allows you to define the structure and presentation of elements on a web page.

CSS: styling structured web content.

Version Control System: Git/GitHub

2.4 RISK MANAGEMENT

Risk is anything that could potentially impact project's timeline, performance, or budget. Risks are potentialities, and in a project management context, if they become realities, they then become classified as "issues" that must be addressed.

The risk in our case is not having clean data so that our accurac y won't be there and cleaning data is never end but fortunately as we used iterative model, we go through each stage many times and each time we minimized the risk of having not clean data as much as we possible

2.5 TESTING

The iterative model supports proactive risk management by addressing potential risks early in the development process. Due to the use of the iterative model after many stages when we rich to the end of the software life cycle stage then we test the accuracy. if it's not convincing then we did another iterative and this goes on till we got a good result

And testing tool including:

- API testing with postman tool
- Confusion matrix is a table that is used to evaluate the performance of a classification model. It provides a summary of the predictions made by the model compared to the actual ground truth labels.
- Clustering visualization in 2d and 3d forms help in checking overlapping of clutters and depth of each one

3. ANALYSIS AND DESIGN

3.1 Database Design

we create ER diagram of the data how should be stored, and definitely is open for new modifications.

These two columns are for users and authentication functionalities that will include login and register

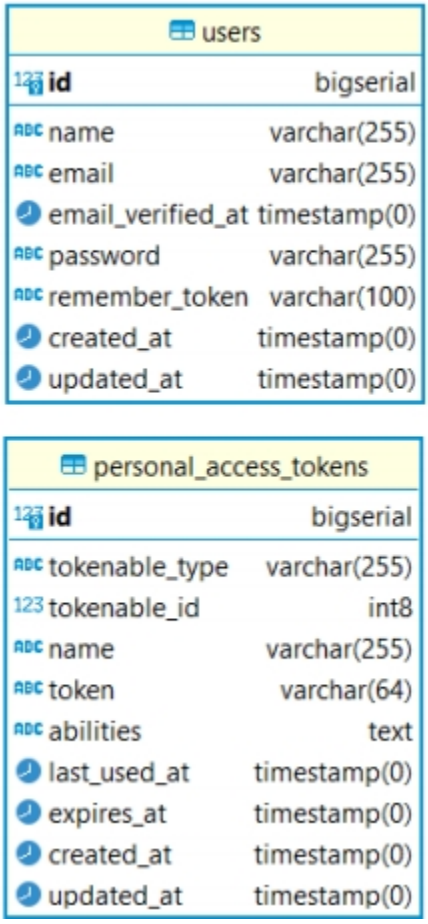

Figure 2 ER - user authentication related table

As clustering is a part of a bigger task called Takhrij (التخريج), the hadiths should be connected to subchapters, chapters, and books that include them. Takhrij involves organizing the hadiths in a specific order based on their numbers in the book

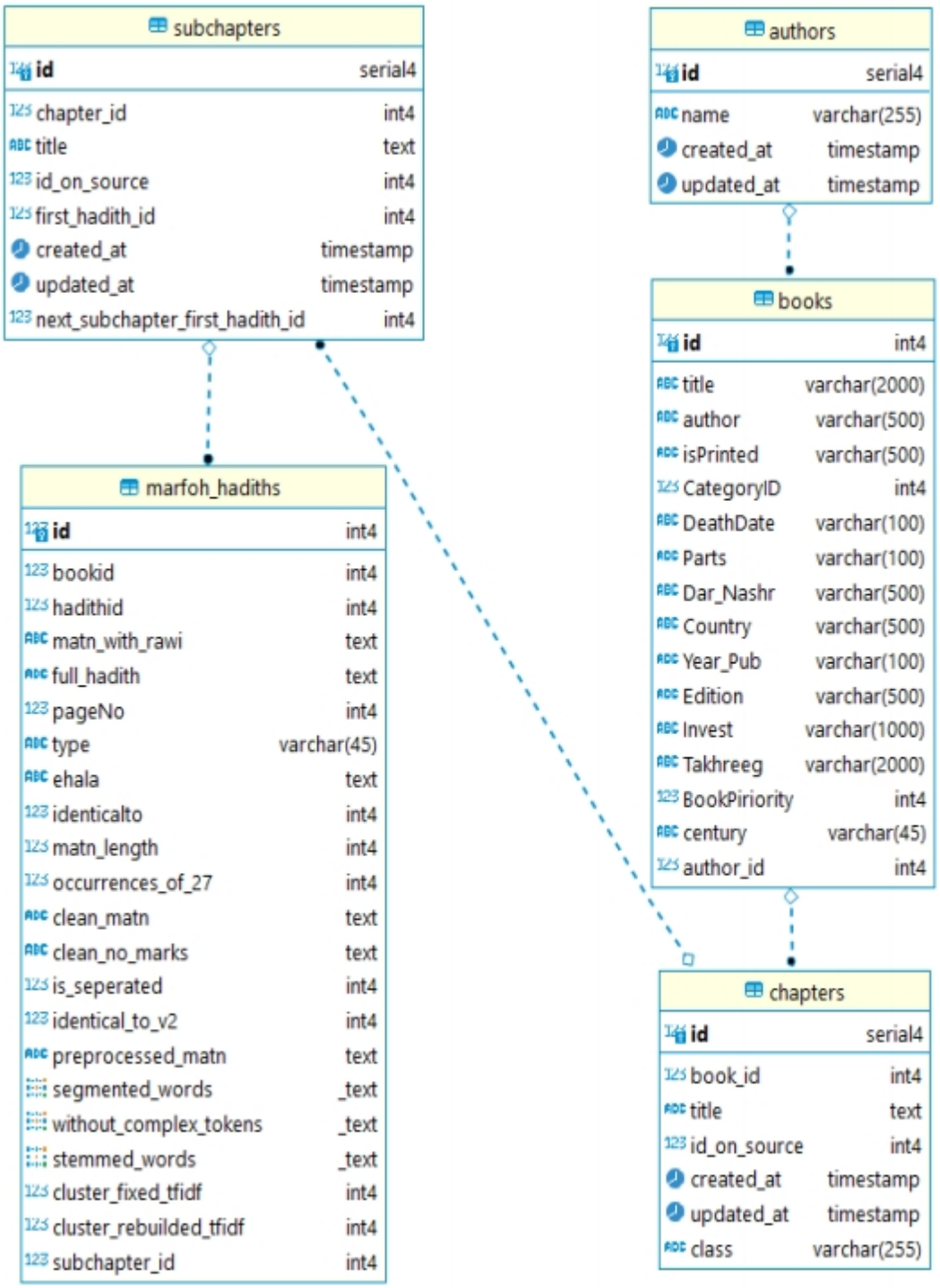

Figure 3 ER -relations of books hadiths related tables

those columns on left side are part of marfoh_hadiths table as could see in previous page, they being populated with Matn contains after being cleaned and preprocessed. And columns store result of different cluster algorithms.

clean_matn	text
clean_no_marks	text
is_seperated	int4
identical_to_v2	int4
preprocessed_matn	text
segmented_words	_text
without_complex_tokens	_text
stemmed_words	_text
cluster_fixed_tfidf	int4
cluster_rebuilded_tfidf	int4

Figure 4 ER columns of preprocessed data

those tables being used in clean and preprocessing functions. words: is table that contains about 600,000 words with its root. Segmented_words: is table that contain words being segmented, segmenting means deleting suffix and prefix of word and. this table contain also stem of those Segmented words.

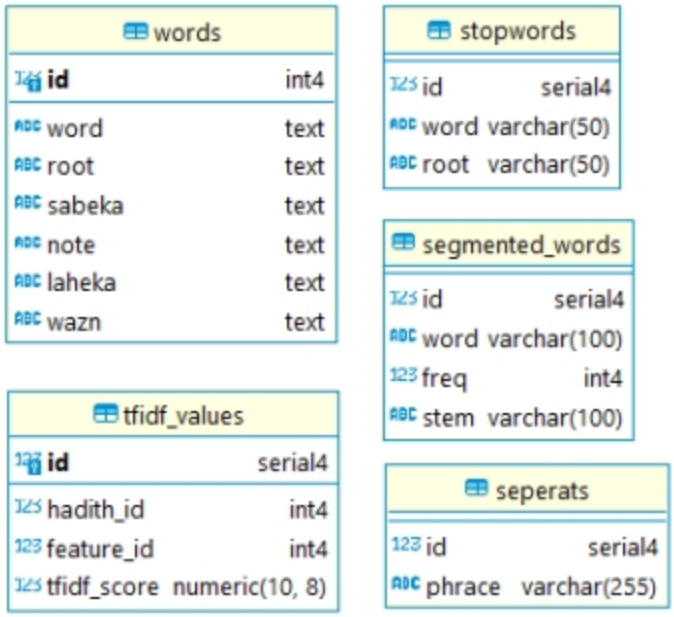

Figure 5 ER tables of word and phrases need for preprocessing data

3.2 clustering/classification software implementation life cycle

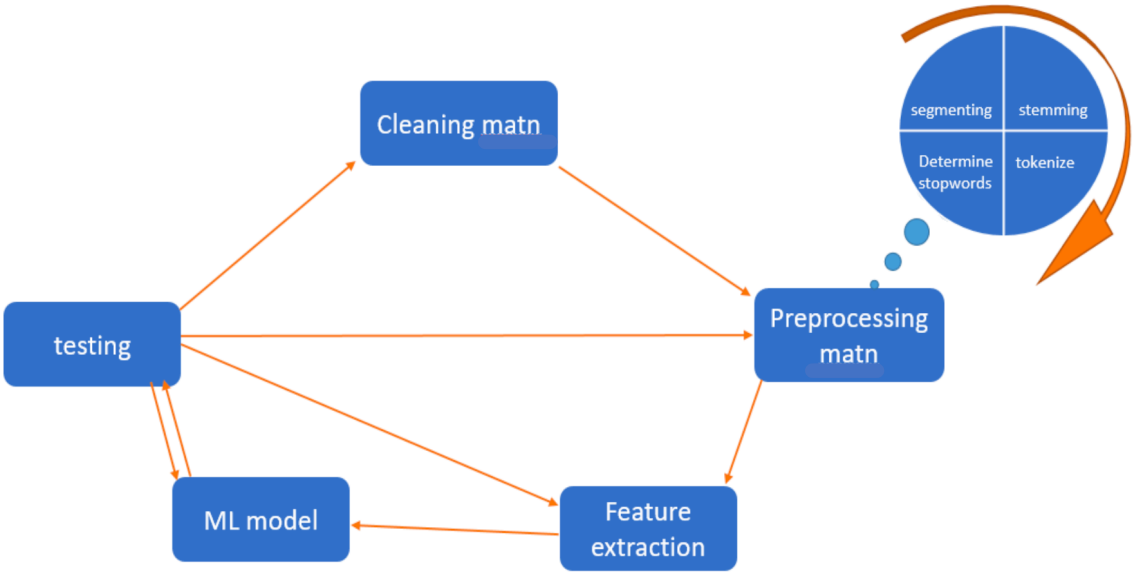

Figure 6 clustering/classification software implementation life cycle

The graph above illustrates the general life cycle of processing Hadith Matn data. The process begins with the cleaning phase, which involves separating the Matn from the Sanad and removing any comments made by narrators. Following this, the Matn undergoes preprocessing steps such as stop word removal, segmentation, and stemming. Once preprocessed, the Matn is tokenized, breaking it down into individual units.

The next step involves extracting numeric features from the Matn using the term frequency-inverse document frequency (TF-IDF) technique. These features represent the importance of words in the Matn relative to the entire dataset.

Once the Matn is prepared with its features, it is fed into a machine learning model for further analysis and processing. The model leverages the extracted features to learn patterns and make predictions based on the given data.

Finally, the cycle concludes with testing the results obtained from the machine learning model. Based on the understanding of the results, adjustments can be made by either revisiting the initial cleaning phase or directly proceeding to the third preprocessing step, depending on the specific requirements and observations. This iterative process allows for refining and optimizing the analysis of Hadith Matn data.

3.3 TF-IDF equation

One of the fundamental equations in information retrieval and text analysis is the term frequency-inverse document frequency (TF-IDF) equation. TF-IDF captures the relevance of terms within a document collection. It combines term frequency (TF) and inverse document frequency (IDF) to calculate the importance of a term in a document relative to the entire corpus. The equation for calculating TF-IDF is as follows:

TF-IDF (t, d) = TF (t, d) * IDF(t)

Where:

- TF (t, d) represents the term frequency of term t in document d.
- IDF(t) denotes the inverse document frequency of term t across the document collection.

The TF component measures the frequency of a term within a specific document, providing insight into its local importance. It is typically calculated as the ratio of the number of times a term occurs in a document to the total number of terms in that document.

The IDF component evaluates the overall importance of a term across the entire document collection. It quantifies how frequently a term occurs in the corpus and penalizes frequently occurring terms. IDF is often computed as the logarithm of the ratio between the total number of documents in the collection and the number of documents containing the term.

The multiplication of TF and IDF yields the TF-IDF score, which indicates the significance of a term within a document and across the collection. Higher TF-IDF scores suggest greater importance and relevance of a term to a particular document.

The TF-IDF equation has wide-ranging applications in information retrieval and natural language processing. It is frequently used in tasks such as document ranking, text classification, keyword extraction, text clustering, and more. By incorporating TF-IDF, researchers and practitioners can effectively measure the importance of terms, improve document retrieval accuracy, and enhance various text analysis and information retrieval tasks.

Understanding the TF-IDF equation and its significance is crucial for developing robust information retrieval systems and performing advanced text analysis. By leveraging TF-IDF, researchers and practitioners can extract valuable insights from document collections and effectively process textual data. (Christopher , et al., 2008)

4. RESULTS AND DISCUSSIONS

This chapter presents the results and discussions of the project, focusing on clustering and classification outcomes. Additionally, it includes the reusable software for implementation of cleaning algorithms, stop words management, and preprocessing techniques for the stored data. The results are organized and presented through tables, figures, and curves, providing a concise overview of the findings. Following the presentation of results, a critical and comprehensive discussion is provided to interpret and analyze the outcomes in detail.

4.1 Matn Cleaning Software

Reusable Clean cleaning software will provide pre-built functions and algorithms that can be easily integrated into the data preprocessing pipeline. This significantly reduces the time and effort required for manual data cleaning, allowing researchers and practitioners to focus more on the clustering and classification tasks.

Hadith	140 /2 /21272 $53657أَخْبَرَنَاهُ 94/أَبُو الْقَاسِمِ بْنُ السَّمَرْقَنْدِيِّ 94/ L12355، أَنْبَأَنَا 26/أَبُو مُحَمَّدِ ابْنُ أَبِي عُثْمَانَ 26/ L62872، 26/وَأَبُو طَاهِرٍ الْقَصَّارِيُّ . 26/ L65195 ح وأَخْبَرَنَا 94/أَبُو عَبْدِ اللَّهِ بْنُ الْقَصَّارِيِّ 94/L80792، أَنْبَأَنَا 26/أَبِي 26/ L65195، قَالا: أَنْبَأَنَا 26/أَبُو الْقَاسِمِ الصَّرْصَرِيُّ . 26/ L12304ح وَأَخْبَرَنَا 94/أَبُو مَنْصُورِ بْنُ الرَّزَّازِ 94/ L78251، 94/وَأَبُو الطَّيِّبِ الْكُتَامِيُّ 94/ L80795، 94/وَأَبُو الْحَسَنِ الأَنْدَلُسِيُّ 94/ L46378، 94/وَالْخَيَّاطُ 94/ L82798، 94/وَأَبُو الْبَيْضَاءِ الْحَجَبِيُّ L80796 94/، 94/وَأُمُّ الْحَسَنِ 94/ L86476، قَالُوا: أَنْبَأَنَا 26/أَبُو الْخَطَّابِ 26/ L46427، أَنْبَأَنَا 26/أَبُو مُحَمَّدٍ الْبَيِّعُ 26/ L22556، قَالا: حَدَّثَنَا 26/الْحُسَيْنُ بْنُ إِسْمَاعِيلَ L15253 26/، حَدَّثَنَا 26/عَلِيُّ بْنُ مُسْلِمٍ 26/ L5815، حَدَّثَنَا 26/مُحَمَّدُ بْنُ يَزِيدَ الْوَاسِطِيُّ L7331 26/، حَدَّثَنَا 26/عَاصِمُ بْنُ رَجَاءٍ 26/ L4062، عَنْ 26/كَثِيرِ بن قيس 26/ L6572، 27/عَنْ 93/أَبِي الدَّرْدَاءِ 93/ L6271قَالَ: جَاءَهُ رَجُلٌ مِنْ أَهْلِ الْمَدِينَةِ ، وَهُوَ بِمِصْرَ ، قَالَ: فَقَالَ لَهُ أَبُو الدَّرْدَاءِ: مَا أَقْدَمَكَ أَيْ أَخِي ؟ قَالَ: حَدِيثٌ بَلَغَنِي عَنْكَ أَنَّكَ تُحَدِّثُ بِهِ عَنْ رَسُولِ اللَّهِ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ ، قَالَ: أَمَا قَدِمْتَ لِتِجَارَةٍ ؟ قَالَ: لا ، قَالَ: أَمَا قَدِمْتَ لِطَلَبِ حَاجَةٍ ؟ قَالَ: لا ، قَالَ: فَمَا قَدِمْتَ إِلا لِطَلَبِ -وَقَالَ الصَّرْصَرِيُّ: فِي طَلَبِ - هَذَا الْحَدِيثِ ؟ قَالَ: فَإِنِّي سَمِعْتُ رَسُولَ اللَّهِ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ يَقُولُ " 20/: مَنْ سَلَكَ طَرِيقًا يَطْلُبُ فِيهِ عِلْمًا سَلَكَ اللَّهُ بِهِ طَرِيقًا إِلَى الْجَنَّةِ ، وَإِنَّ 57/ الْمَلائِكَةَ 57/ لَتَضَعُ أَجْنِحَتَهَا رِضًا لِطَالِبِ الْعِلْمِ ، وَإِنَّ الْعَالِمَ يَسْتَغْفِرُ لَهُ مَنْ فِي السَّمَوَاتِ وَالأَرْضِ حَتَّى الْحِيتَانُ جَوْفَ - /23 وَقَالَ الصَّرْصَرِيُّ : فِي جَوْفِ - /23 الْمَاءِ ، وَلِفَضْلِ الْعَالِمِ عَلَى الْعَابِدِ /23 - وَقَالَ أَبُو مُحَمَّدٍ : عَلَى الْعَامِلِ /23 - كَفَضْلِ الْقَمَرِ عَلَى سَائِرِ الْكَوَاكِبِ ، إِنَّ الْعُلَمَاءَ وَرَثَةُ الأَنْبِيَاءِ ، إِنَّ الأَنْبِيَاءَ لَمْ يُوَرِّثُوا دِينَارًا وَلا دِرْهَمًا ، وَإِنَّمَا وَرَّثُوا الْعِلْمَ ، فَمَنْ أَخَذَ بِهِ أَخَذَ بِحَظٍّ وَافِرٍ . 27/ "وَرَوَاهُ أَحْمَدُ بْنُ حَنْبَلٍ ، عَنْ مُحَمَّدِ بْنِ يَزِيدَ هَكَذَا ، إلا أَنَّهُ قَلَبَهُ ، فَقَالَ: قَيْسُ بْنُ كَثِيرٍ * .
Cleaned matn	من سلك طريقا يطلب فيه علما سلك الله به طريقا إلى الجنة وإن الملائكة لتضع أجنحتها رضا لطالب العلم وإن العالم يستغفر له من في السموات والأرض حتى الحيتان جوف الماء ولفضل العالم على العابد كفضل القمر على سائر الكواكب إن العلماء ورثة الأنبياء إن الأنبياء لم يورثوا دينارا ولا درهما وإنما ورثوا العلم فمن أخذ به أخذ بحظ وافر

4.2 stopwords management software

This reusable software facilitates the insertion, update, and deletion of stopwords. Determining stopwords can be challenging as it sometimes requires adding all forms of a word as stopwords. This software performs web scraping on https://tasrif.reverso.net/ to fetch all word forms, if needed.

For example, the word 'ســـمع' should not be considered for clustering or classification, but it has numerous forms. Manually determining and adding each form to the stopwords list would be a difficult task."

Figure 7 word forms of سمع

4.3 tokenization software

This reusable software will do tokenization including those phases

- prefix segmenting feature of word ex.: فاسمعوا => سمعوا
- remove stopwords
- manage complex words that include more than one word in Arabic names ex:
  - علي بن أبي طالب
  - عبد الرحمن بن عوف
- Word stemming ex: سمعوا => سمع

Cleaned Matn:

صدق فقال النبي صلى الله عليه وسلم انتقلي إلى منزل ابن أم مكتوم فاعتدي عندها ثم قال لا أم كلثوم يكثر عوادها ولكن انتقلي إلى عبد الله بن أم مكتوم فإنه أعمى فانتقلت إلى عبد الله فاعتدت عنده حتى انقضت عدتها ثم خطبها أبو جهم ومعاوية بن أبي سفيان فجاءت رسول الله صلى الله عليه وسلم تستأمره فيهما فقال أبو جهم أخاف عليك قسقاسته للعصا وأما معاوية فرجل أخلق من المال فتزوجت أسامة بن زيد بعد ذلك

Segmented tokenize:

صدق	انتقلي	منزل	اعتدي
"ابن أم مكتوم"	"أم كلثوم"	يكثر	عوادها
انتقلي	"عبد الله بن أم مكتوم"	أعمى	انتقلت
عبد	الله	اعتدت	انقضت
عدتها	خطبها	"أبو جهم"	"معاوية بن أبي سفيان"
جاءت	"رسول الله"	تستأمره	"أبو جهم"
أخاف	قسقاسته	عصا	معاوية
رجل	أخلق	مال	تزوجت
"أسامة بن زيد"			

Stemmed tokenize:

صدق	نقل	نزل	عدد
كثر	عود	نقل	عمي
نقل	عبد	الله	عدا
قضض	عدد	خطب	جيأ
أمر	خوف	قسس	عصا
معاوية	رجل	خلق	مول
زوج			

This list of stemmed words, without specific stopwords identified, will be used as input for feature extraction.

4.3 feature extraction software

the results of applying the TF-IDF technique to hadiths are presented. The results are organized and saved in database tfidf_values table, is here being figured the first top 200 words by text-cloud representation. The TF-IDF scores reveal the importance of different terms within the texts, indicating their relevance and significance. The discussion critically analyzes the findings, highlighting the connections between key terms and the overall themes of the texts

Figure 8 top 200 words result of tf-idf

4.4 clustering result

the results of the clustering analysis are presented, accompanied by visualizations that aid in evaluating the outcomes. The clustering algorithm was applied to the dataset, grouping similar data points together based on their features and patterns. The results are showcased through clustering visualizations such as scatter plots, allowing for a clear and intuitive representation of the clustering structure. These visualizations facilitate the evaluation of the clustering results by providing insights into the separation, compactness, and coherence of the identified clusters. The discussion critically examines the quality and effectiveness of the clustering outcomes, considering factors such as cluster cohesion, separation, and interpretability. The clustering visualizations serve as a valuable tool for assessing the performance of the algorithm, validating the clustering results, and gaining a deeper understanding of the underlying data patterns and relationships.

In result number of hadith being clustered is 300,000 within 13,000 clusters as accuracy rate more about 95%

This diagram only show resulted 22 clustered for evaluating specific range of clustered hadiths

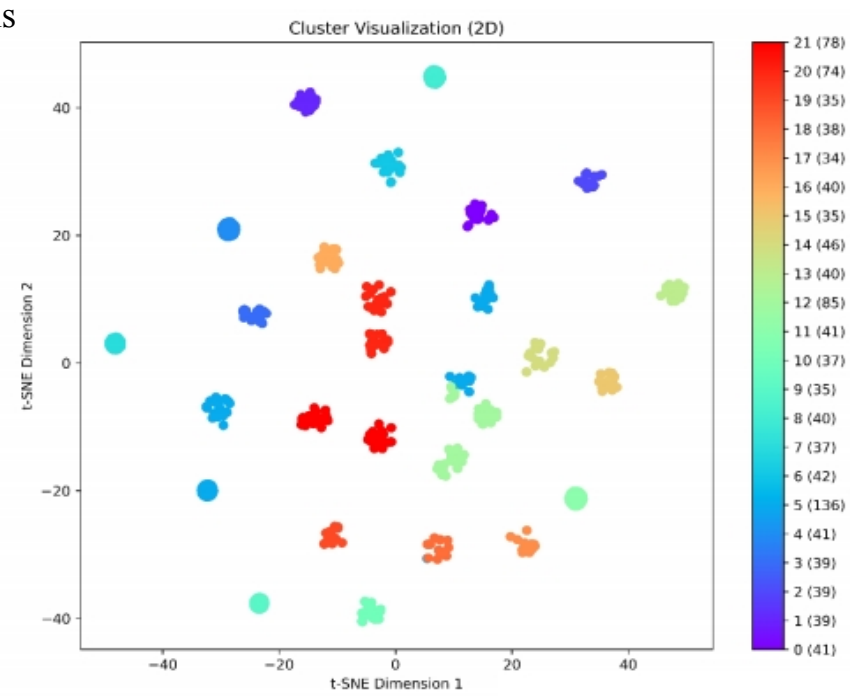

Figure 9 2d-clustering for 22 cluster and 1000 hadiths

As two-dimensional (2D) clustering visualization is effective for examining separated or overlapped clusters. On the other hand, 3D clustering visualization offers main advantages for inspecting the similarity of individual elements within clusters. The proximity of data points and the depth of clusters can provide valuable insights for evaluating the quality and structure of the clustering results. By utilizing the additional dimension, 3D clustering visualization allows for a deeper analysis of the relationships and patterns within each cluster, enhancing the evaluation process.

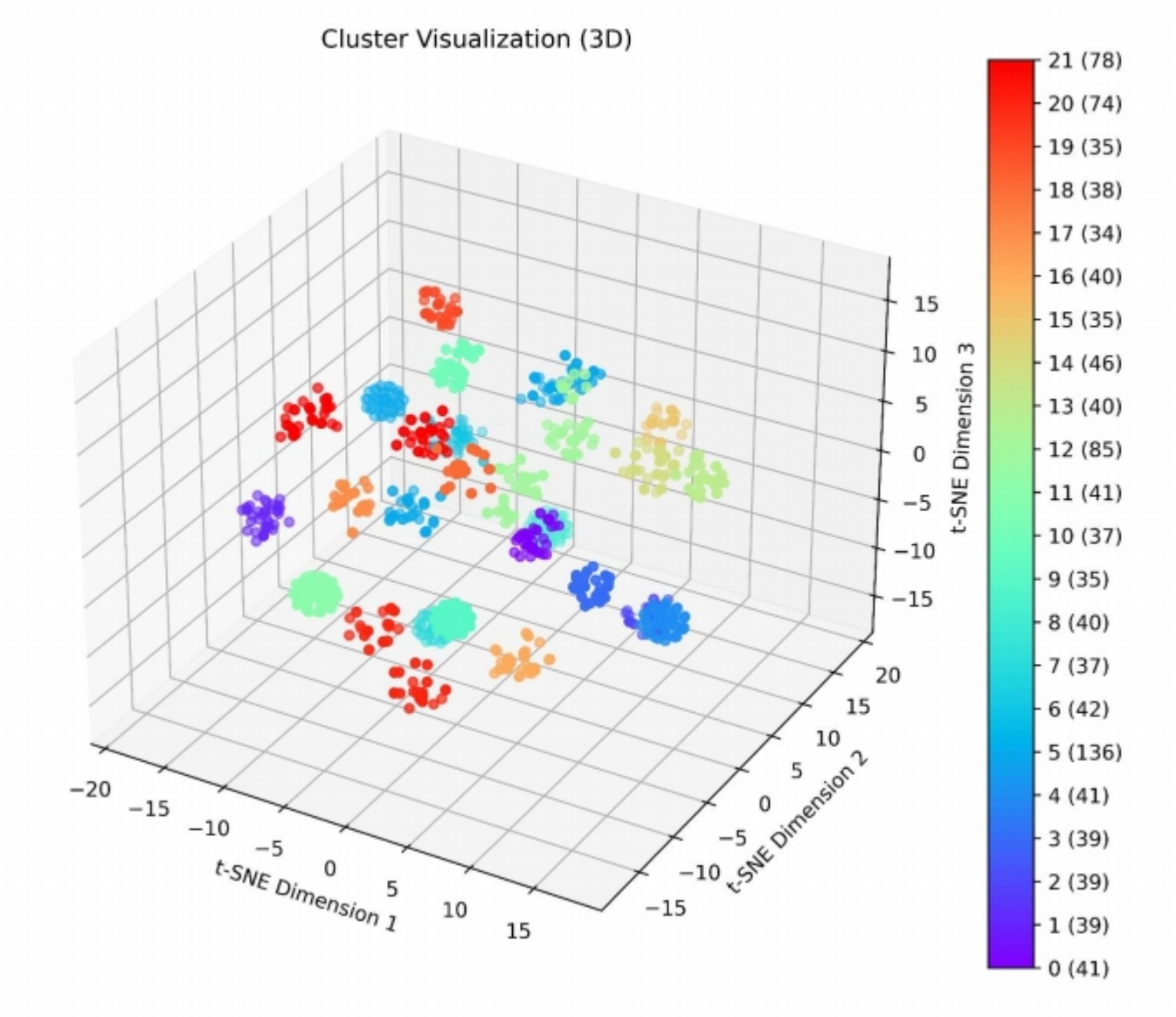

Figure 10 3d-clustering for 22 cluster and 1000 hadiths

4.5 Classification result

In our text classification task using Logistic Regression, we observed a significant improvement in the results, with accuracy increasing from 46% to 70%. However, further enhancements are still needed. To address this, we plan to review the data cleaning process and iterate through preprocessing steps and feature extraction techniques. One challenge we encountered is the presence of hadiths containing contextual information that could belong to multiple classes, introducing ambiguity in classification. To gain deeper insights into the performance, we utilized a confusion matrix to showcase the classification results for each class. This analysis will guide us in refining the classification model and addressing the challenges associated with multi-class classification and context-aware data.

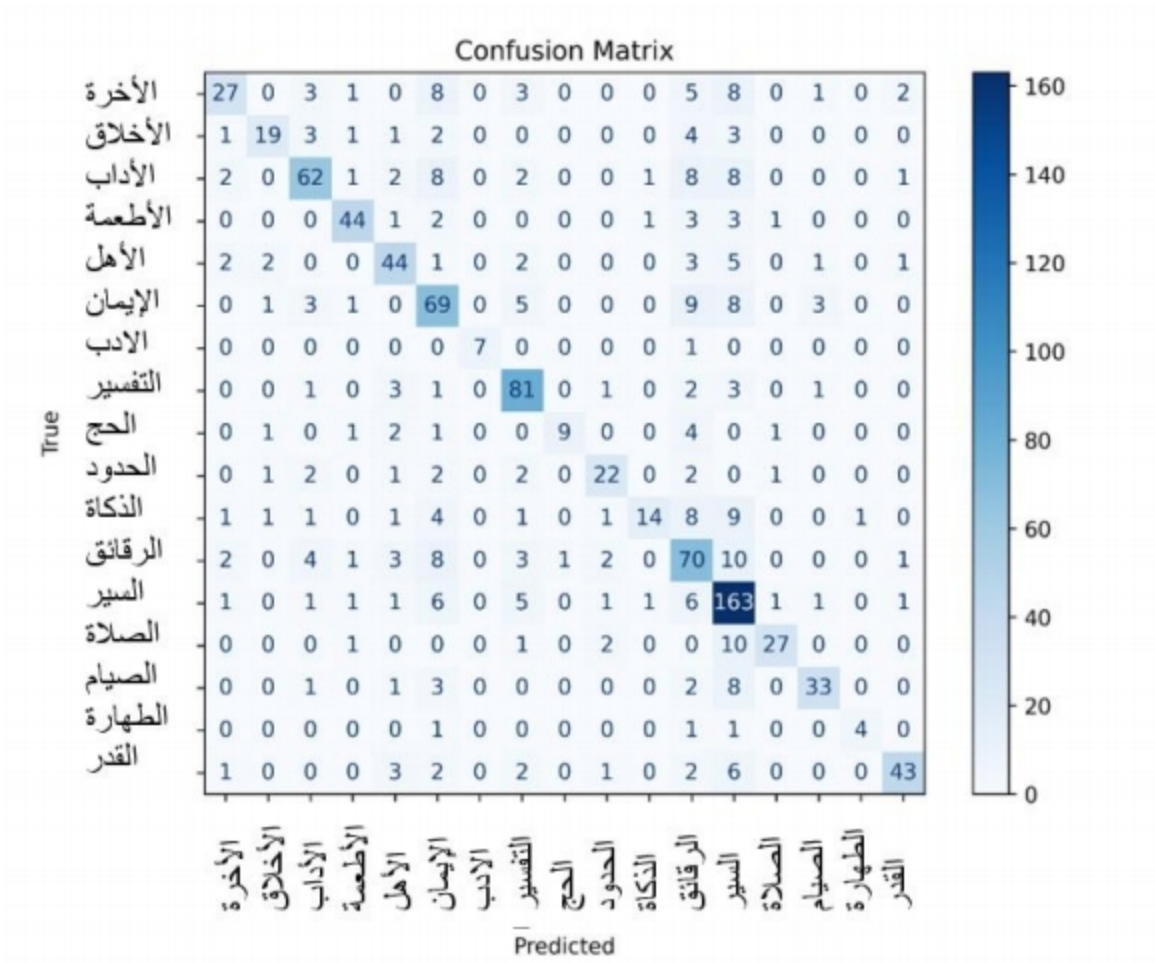

Figure 11 confusion marix

4.6 HADITHS CHECKING WEB APP

As part of the Takhrij process, which involves reviewing hadiths and extracting Shawahid (supporting evidence), a web application is currently being developed. This application aims to provide hadith specialists with a platform that facilitates their interaction with the hadiths, granting them authenticated access and the ability to make modifications as necessary. The web application will streamline the Takhrij process, allowing specialists to efficiently manage and analyze the hadiths while ensuring their authenticity and accuracy.

4.6.1 web app security

As enabling hadith modification should only be for authenticated users. It's being need for functionalities to facilities that as providing login screen.it uses session-based authentication that in Laravel framework enabling CSRF protection.

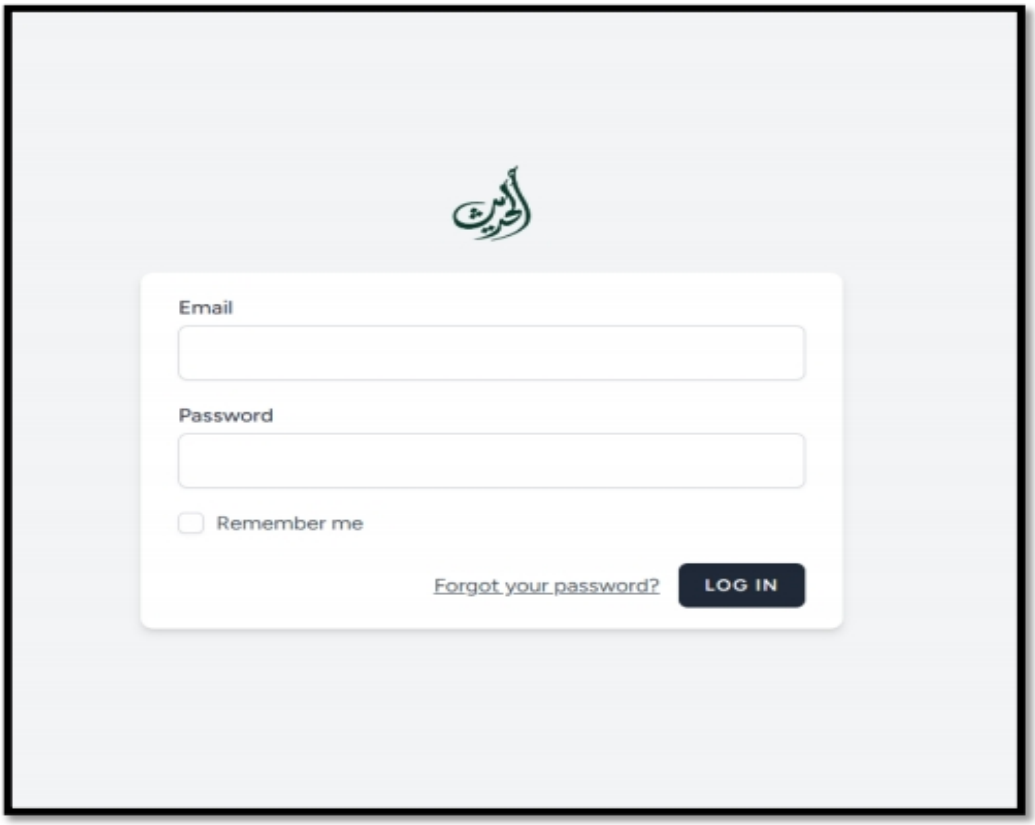

Figure 12 authentication screen

4.6.2 web app tasks screen

After login will be routes directly to tasks screen. From here he will select his screen that allowed him to enter.

The tasks included in the web app are as follows:

- Separating the Matn (content) from the Sanad (chain of narrators) and extracting any comments or annotations from scholars.
- Verifying the position of the hadith within the book, including its order number.
- Checking the sequence of annotators or narrators of the hadiths, ensuring the correct order is maintained.

Figure 13 web app tasks screen

4.6.3 web app select book screen

All the tasks after being selected will be directed to this screen selecting book and hadith number before go to task screen itself.

In select book input field had feature of search in drop down menu to facilitate the process as we have about 1000 books

Figure 14 web app select book and hadith number screen

4.6.4 web app Matn checking screen

This particular screen serves a crucial purpose by facilitating the meticulous examination of Hadith texts. Its primary function involves scrutinizing the Matn (textual content) of a Hadith, ensuring its accuracy and authenticity. However, its capabilities extend beyond this fundamental aspect.

In addition to verifying the Matn, this screen undertakes a comprehensive comparison between the entirety of a given Hadith and the corresponding Hadith in its original book, which is conveniently available in PDF format. By leveraging this innovative feature, scholars and researchers can perform a detailed analysis to identify any potential discrepancies, variations, or deviations that may have occurred during the transmission of the Hadith.

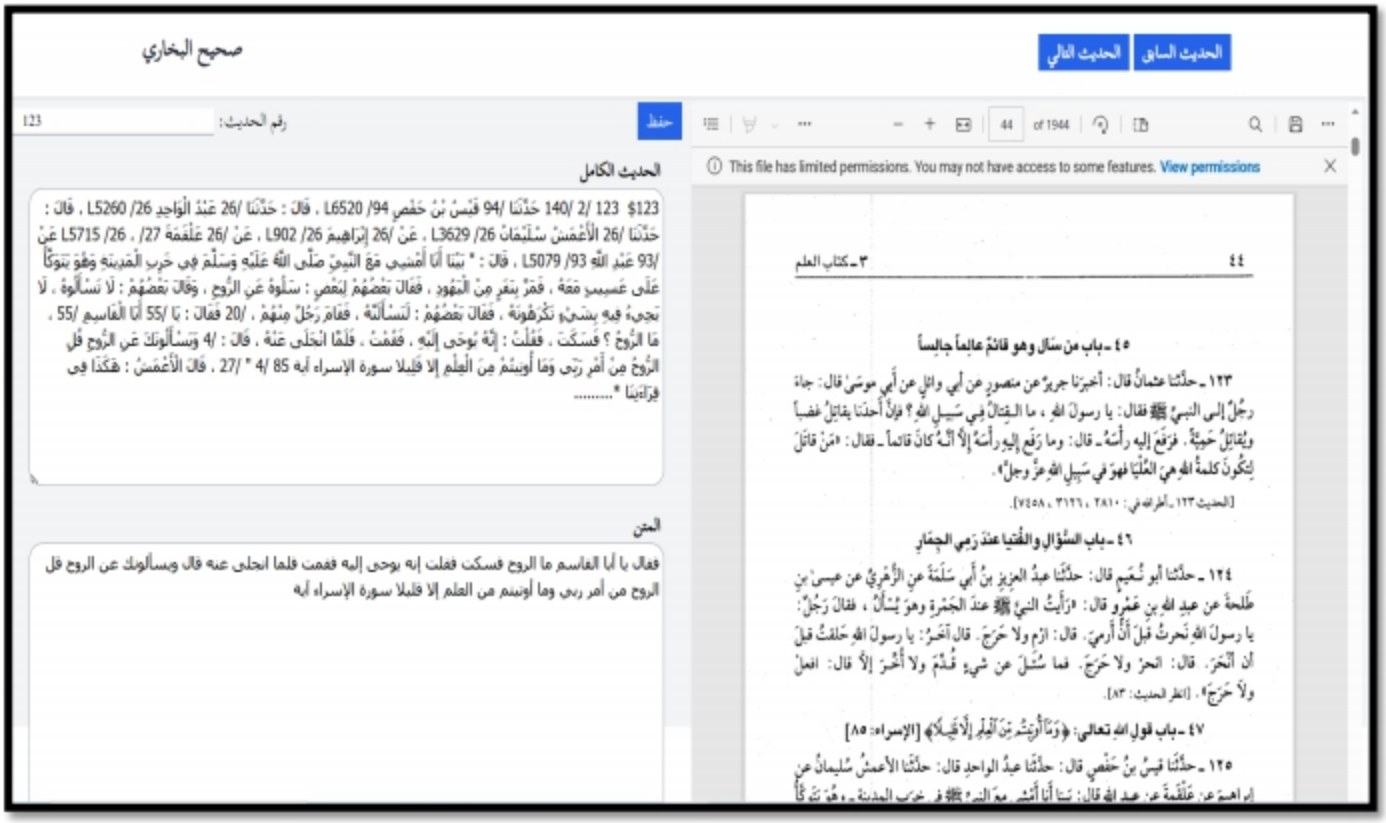

Figure 15 web app check matn task screen

5. CONCLUSION

The completion of this project marks an important milestone in the study of Hadith Matns. Our primary objective was to analyze these texts and employ clustering and classification techniques to identify and extract Shawahid (evidence) and conduct Takhrij of Hadiths. We started by preparing the texts, separating the main content from other parts, and cleaning them up. This involved breaking the texts into smaller pieces, removing unnecessary words, and reducing words to their basic form.

Once the texts were ready, we focused on extracting important features using a method called TF-IDF. This technique helped us identify key words and concepts that distinguish one Hadith from another. With these features in hand, we applied artificial intelligence models to cluster and classify the Hadiths, finding similarities and organizing them into groups.

Our primary focus for future is to enhance the accuracy of the clustering and classification processes. Recognizing the importance of collaboration, we seek to actively engage with esteemed Hadith scholars. By involving their invaluable expertise and insights, we anticipate refining the cleaning phase to render it even more tailored to the analysis of Hadith texts. Their guidance will help us address potential challenges and nuances associated with the preprocessing of data, enabling us to achieve more precise and reliable results.

Furthermore, as technology advances, we are open to exploring alternative approaches to feature extraction. One potential avenue involves merging TF-IDF with word embedding techniques. This amalgamation could offer a more comprehensive representation of the semantic meaning and context within the Hadith texts. By integrating different tech tools and methodologies, we aim to enhance the overall efficiency and accuracy of our analysis.

This project represents an ongoing journey, characterized by continuous learning and adaptation. We are keenly aware that studying Hadith Matns requires unwavering dedication and ongoing effort. By fostering collaboration, embracing new technologies, and persistently seeking knowledge, we are committed to expanding our understanding and appreciation of this significant religious tradition.

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